

Supplementary Material for “Leakage-Aware Multi-Horizon Benchmarking of Compact Neural PV Forecasts Across Co-Located Technologies”

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CONFIRMATION

1 Evidence scope

This supplement reports the frozen evidence supporting the main manuscript. The benchmark comprises four compact neural implementations, three co-located photovoltaic arrays, three random seeds (42, 43, and 44), and direct 144-step forecasts at five-minute resolution. Primary H12, H48, H96, and H144 metrics use horizon-specific valid origins. Daily Persistence comparisons instead use an elementwise intersection mask that is valid for the daily lag, labels, Last-value Persistence, and every neural forecast.

2 Input and preprocessing configuration

Each history has 72 time steps and 17 channels: Performance Ratio; ambient temperature; relative humidity; global and diffuse horizontal radiation; global and diffuse tilted radiation; Active Power; eight corresponding missingness indicators; and one Isolation Forest indicator. The imputer, feature scaler, target scaler, and Isolation Forest are fitted on Train only. No forecast-origin successor value enters the input. Windows remain within chronological splits and within a regular five-minute coordinate.

Table S1: Fixed optimization configuration for all compact neural implementations.

Alt text: The shared optimization configuration fixes the optimizer, learning rate, batch size, epoch budget, early-stopping rule, gradient clipping, H144 validation objective, and three random seeds for all four compact implementations.

Setting	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-5}
Batch size	256
Maximum epochs	25
Early-stopping patience	5
Minimum improvement	10^{-8}
Gradient clipping	1.0
Checkpoint objective	Validation global masked MSE on H144
Random seeds	42, 43, 44

3 Architecture identity and full layer configuration

The Discrete recurrent decoder combines three temporal-convolution branches (dilations 1, 2, and 4; 24 channels each), a 64-dimensional one-layer Transformer, a learned feature gate, bidirectional GRU attention, and an autoregressive GRUCell trajectory decoder. The Inverted-variate Transformer treats the 17 variables as tokens, projects each 72-point history to 64 dimensions, applies one four-head Transformer layer, then flattens the encoded tokens into a 144-output linear map. The Joint-patch Transformer forms overlapping joint multivariate patches of length 12 and stride 6, uses one 64-dimensional four-head Transformer layer, and maps the encoded patches to 144 outputs. The Depthwise TCN uses an input projection followed by four depthwise-separable temporal blocks with 64 channels and kernel size 5, then a flattened linear H144 head.

These are project implementations for a controlled mechanism comparison. The inverted model omits the full official iTransformer embedding, normalization, and decoder design; the joint-patch model does not use the channel-independent PatchTST pipeline; and the depthwise model omits the full ModernTCN large-kernel and multi-scale block system. Accordingly, the reported rankings do not rank the official architectures.

4 Metrics and masks

For a fixed valid-point set $\mathcal{I}$ with $n = |\mathcal{I}|$,

$$\text{RMSE} = \sqrt{n^{-1} \sum_{i \in \mathcal{I}} (\hat{y}_i - y_i)^2}, \quad (\text{S1})$$

$$\text{MAE} = n^{-1} \sum_{i \in \mathcal{I}} |\hat{y}_i - y_i|, \quad (\text{S2})$$

$$R^2 = 1 - \frac{\sum_{i \in \mathcal{I}} (\hat{y}_i - y_i)^2}{\sum_{i \in \mathcal{I}} (y_i - \bar{y})^2}, \quad (\text{S3})$$

$$\text{range-nRMSE} = \frac{\text{RMSE}}{y_{\max, \text{Train}} - y_{\min, \text{Train}}}, \quad (\text{S4})$$

$$\text{RMSE skill} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{reference}}}. \quad (\text{S5})$$

The daylight mask is true target Active Power greater than 1% of the corresponding array’s Train maximum. It is an evaluation-only mask. Daily Persistence takes each target timestamp’s power 24 hours earlier without interpolation. If that point is unavailable, it is removed from every method in that Daily-matched comparison.

5 Complete quantitative evidence

The following tables are generated directly from the frozen `corrected_metrics.csv`; they are not transcribed into this source file. They provide per-seed primary metrics, primary support, complete-H144-origin sensitivity, and matched Daily Persistence results.

Table S2: Primary per-seed RMSE values.

Alt text: Per-seed root mean square errors show how forecast accuracy changes across arrays, scopes, and horizons for the four compact neural implementations. Seed spread reflects initialization variability rather than independent data replication.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.20699	0.21172	0.20282
Sanyo	Full	H12	Inverted-variate	0.26513	0.27584	0.22716
Sanyo	Full	H12	Joint-patch	0.29326	0.24817	0.21943
Sanyo	Full	H12	Depthwise TCN	0.23350	0.17550	0.25439
Sanyo	Daylight	H12	Discrete recurrent	0.29601	0.30675	0.29114
Sanyo	Daylight	H12	Inverted-variate	0.31789	0.32546	0.30911
Sanyo	Daylight	H12	Joint-patch	0.37815	0.32296	0.27677
Sanyo	Daylight	H12	Depthwise TCN	0.31123	0.24041	0.32051
Sanyo	Full	H48	Discrete recurrent	0.64683	0.64400	0.66629
Sanyo	Full	H48	Inverted-variate	0.68041	0.62303	0.62109
Sanyo	Full	H48	Joint-patch	0.66859	0.58335	0.59367
Sanyo	Full	H48	Depthwise TCN	0.67710	0.64485	0.63471
Sanyo	Daylight	H48	Discrete recurrent	0.94975	0.94651	0.97978
Sanyo	Daylight	H48	Inverted-variate	0.95042	0.81287	0.85809
Sanyo	Daylight	H48	Joint-patch	0.95240	0.80115	0.83710
Sanyo	Daylight	H48	Depthwise TCN	0.94231	0.90116	0.87112
Sanyo	Full	H96	Discrete recurrent	0.90840	0.90011	0.96696
Sanyo	Full	H96	Inverted-variate	0.85729	0.80883	0.81027
Sanyo	Full	H96	Joint-patch	0.96456	0.84964	0.81070
Sanyo	Full	H96	Depthwise TCN	0.96364	0.85757	0.88544
Sanyo	Daylight	H96	Discrete recurrent	1.34860	1.33368	1.43813
Sanyo	Daylight	H96	Inverted-variate	1.19297	1.08960	1.11853
Sanyo	Daylight	H96	Joint-patch	1.39986	1.20066	1.14794
Sanyo	Daylight	H96	Depthwise TCN	1.36564	1.19882	1.24128
Sanyo	Full	H144	Discrete recurrent	0.87822	0.82591	0.89962
Sanyo	Full	H144	Inverted-variate	0.78097	0.76385	0.78197
Sanyo	Full	H144	Joint-patch	0.84637	0.78215	0.75005
Sanyo	Full	H144	Depthwise TCN	0.85780	0.79629	0.79624
Sanyo	Daylight	H144	Discrete recurrent	1.31199	1.22879	1.34119
Sanyo	Daylight	H144	Inverted-variate	1.07005	1.02634	1.07535
Sanyo	Daylight	H144	Joint-patch	1.22156	1.10516	1.06103
Sanyo	Daylight	H144	Depthwise TCN	1.20759	1.10868	1.11171
Hanwha	Full	H12	Discrete recurrent	0.19283	0.20984	0.23773
Hanwha	Full	H12	Inverted-variate	0.19326	0.21233	0.21254
Hanwha	Full	H12	Joint-patch	0.25045	0.29434	0.23248
Hanwha	Full	H12	Depthwise TCN	0.17950	0.16762	0.17960
Hanwha	Daylight	H12	Discrete recurrent	0.27411	0.30308	0.34010
Hanwha	Daylight	H12	Inverted-variate	0.25729	0.28560	0.28441
Hanwha	Daylight	H12	Joint-patch	0.31341	0.31390	0.28217
Hanwha	Daylight	H12	Depthwise TCN	0.23343	0.22040	0.23794
Hanwha	Full	H48	Discrete recurrent	0.57998	0.57396	0.63790
Hanwha	Full	H48	Inverted-variate	0.54741	0.58955	0.58646
Hanwha	Full	H48	Joint-patch	0.61852	0.57368	0.56605
Hanwha	Full	H48	Depthwise TCN	0.57787	0.58575	0.59752

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Hanwha	Daylight	H48	Discrete recurrent	0.84718	0.84454	0.93710
Hanwha	Daylight	H48	Inverted-variate	0.78356	0.84091	0.82613
Hanwha	Daylight	H48	Joint-patch	0.88106	0.80086	0.79516
Hanwha	Daylight	H48	Depthwise TCN	0.81332	0.82086	0.84039
Hanwha	Full	H96	Discrete recurrent	0.80824	0.85950	0.90481
Hanwha	Full	H96	Inverted-variate	0.84121	0.85535	0.88964
Hanwha	Full	H96	Joint-patch	0.87878	0.76678	0.78162
Hanwha	Full	H96	Depthwise TCN	0.79720	0.79868	0.81582
Hanwha	Daylight	H96	Discrete recurrent	1.19164	1.27417	1.34138
Hanwha	Daylight	H96	Inverted-variate	1.21196	1.22202	1.26041
Hanwha	Daylight	H96	Joint-patch	1.27278	1.08927	1.11426
Hanwha	Daylight	H96	Depthwise TCN	1.13261	1.12349	1.15324
Hanwha	Full	H144	Discrete recurrent	0.76065	0.78426	0.81177
Hanwha	Full	H144	Inverted-variate	0.78583	0.80309	0.82715
Hanwha	Full	H144	Joint-patch	0.79506	0.71160	0.73154
Hanwha	Full	H144	Depthwise TCN	0.72555	0.75028	0.75361
Hanwha	Daylight	H144	Discrete recurrent	1.12700	1.16621	1.20763
Hanwha	Daylight	H144	Inverted-variate	1.12435	1.13795	1.16060
Hanwha	Daylight	H144	Joint-patch	1.14106	1.00309	1.03867
Hanwha	Daylight	H144	Depthwise TCN	1.01361	1.05240	1.05096
Qcells	Full	H12	Discrete recurrent	0.44262	0.73507	1.59137
Qcells	Full	H12	Inverted-variate	0.30176	0.34935	0.33091
Qcells	Full	H12	Joint-patch	0.52988	0.57476	0.49208
Qcells	Full	H12	Depthwise TCN	0.52598	0.30576	0.41648
Qcells	Daylight	H12	Discrete recurrent	0.64247	1.06714	2.31558
Qcells	Daylight	H12	Inverted-variate	0.42778	0.47506	0.46668
Qcells	Daylight	H12	Joint-patch	0.72858	0.81902	0.69755
Qcells	Daylight	H12	Depthwise TCN	0.75796	0.43460	0.60317
Qcells	Full	H48	Discrete recurrent	0.77406	0.94518	1.83265
Qcells	Full	H48	Inverted-variate	0.79118	0.82091	0.81042
Qcells	Full	H48	Joint-patch	0.93780	1.03256	0.98688
Qcells	Full	H48	Depthwise TCN	1.20653	1.04753	1.04156
Qcells	Daylight	H48	Discrete recurrent	1.12663	1.37313	2.67354
Qcells	Daylight	H48	Inverted-variate	1.14009	1.17767	1.16234
Qcells	Daylight	H48	Joint-patch	1.33819	1.49374	1.42803
Qcells	Daylight	H48	Depthwise TCN	1.75350	1.52143	1.51385
Qcells	Full	H96	Discrete recurrent	1.04609	1.15831	1.55933
Qcells	Full	H96	Inverted-variate	1.15002	1.14625	1.11365
Qcells	Full	H96	Joint-patch	1.18191	1.23303	1.17040
Qcells	Full	H96	Depthwise TCN	1.36166	1.43712	1.27932
Qcells	Daylight	H96	Discrete recurrent	1.54353	1.70080	2.29733
Qcells	Daylight	H96	Inverted-variate	1.67275	1.66186	1.60714
Qcells	Daylight	H96	Joint-patch	1.71311	1.79827	1.70418
Qcells	Daylight	H96	Depthwise TCN	1.99499	2.10755	1.86964
Qcells	Full	H144	Discrete recurrent	0.90764	0.92812	0.88946
Qcells	Full	H144	Inverted-variate	0.84633	0.86845	0.81567
Qcells	Full	H144	Joint-patch	0.87026	0.85503	0.82492
Qcells	Full	H144	Depthwise TCN	0.79967	0.79287	0.77389
Qcells	Daylight	H144	Discrete recurrent	1.36729	1.38531	1.32341

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Daylight	H144	Inverted-variate	1.23444	1.25633	1.17410
Qcells	Daylight	H144	Joint-patch	1.26299	1.25253	1.20653
Qcells	Daylight	H144	Depthwise TCN	1.17205	1.16000	1.12490

Table S3: Primary per-seed MAE values.

Alt text: Per-seed mean absolute errors provide a less tail-sensitive comparison of the four compact neural implementations across arrays, scopes, and horizons. The pattern complements RMSE by reducing the influence of large residuals.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.13583	0.12116	0.12189
Sanyo	Full	H12	Inverted-variate	0.20606	0.20906	0.15958
Sanyo	Full	H12	Joint-patch	0.21315	0.17464	0.16188
Sanyo	Full	H12	Depthwise TCN	0.17238	0.11426	0.20034
Sanyo	Daylight	H12	Discrete recurrent	0.22632	0.21644	0.20251
Sanyo	Daylight	H12	Inverted-variate	0.24672	0.25630	0.24797
Sanyo	Daylight	H12	Joint-patch	0.29948	0.25203	0.21011
Sanyo	Daylight	H12	Depthwise TCN	0.25914	0.17448	0.25712
Sanyo	Full	H48	Discrete recurrent	0.31052	0.26504	0.28877
Sanyo	Full	H48	Inverted-variate	0.41494	0.40296	0.36613
Sanyo	Full	H48	Joint-patch	0.37397	0.33657	0.32167
Sanyo	Full	H48	Depthwise TCN	0.40982	0.37110	0.37789
Sanyo	Daylight	H48	Discrete recurrent	0.59637	0.52821	0.55362
Sanyo	Daylight	H48	Inverted-variate	0.63958	0.52668	0.58159
Sanyo	Daylight	H48	Joint-patch	0.61831	0.49596	0.51232
Sanyo	Daylight	H48	Depthwise TCN	0.64092	0.58876	0.54797
Sanyo	Full	H96	Discrete recurrent	0.45539	0.42600	0.46402
Sanyo	Full	H96	Inverted-variate	0.54810	0.53406	0.50433
Sanyo	Full	H96	Joint-patch	0.55816	0.51419	0.46833
Sanyo	Full	H96	Depthwise TCN	0.61608	0.52564	0.54538
Sanyo	Daylight	H96	Discrete recurrent	0.92720	0.88733	0.95360
Sanyo	Daylight	H96	Inverted-variate	0.88941	0.80380	0.83193
Sanyo	Daylight	H96	Joint-patch	1.00084	0.85190	0.79454
Sanyo	Daylight	H96	Depthwise TCN	1.05117	0.88114	0.89613
Sanyo	Full	H144	Discrete recurrent	0.45801	0.40717	0.46249
Sanyo	Full	H144	Inverted-variate	0.50741	0.51151	0.50399
Sanyo	Full	H144	Joint-patch	0.49204	0.47784	0.44669
Sanyo	Full	H144	Depthwise TCN	0.56495	0.51928	0.51794
Sanyo	Daylight	H144	Discrete recurrent	0.96352	0.86876	0.97403
Sanyo	Daylight	H144	Inverted-variate	0.81015	0.78388	0.83433
Sanyo	Daylight	H144	Joint-patch	0.87534	0.80142	0.76574
Sanyo	Daylight	H144	Depthwise TCN	0.95405	0.87339	0.86283
Hanwha	Full	H12	Discrete recurrent	0.12387	0.12836	0.16202
Hanwha	Full	H12	Inverted-variate	0.13972	0.15415	0.15395
Hanwha	Full	H12	Joint-patch	0.19486	0.24511	0.17962
Hanwha	Full	H12	Depthwise TCN	0.12813	0.11327	0.12656
Hanwha	Daylight	H12	Discrete recurrent	0.20778	0.22419	0.28126
Hanwha	Daylight	H12	Inverted-variate	0.19702	0.22770	0.22425

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Hanwha	Daylight	H12	Joint-patch	0.25158	0.24992	0.22135
Hanwha	Daylight	H12	Depthwise TCN	0.18270	0.15758	0.18161
Hanwha	Full	H48	Discrete recurrent	0.27710	0.23620	0.32083
Hanwha	Full	H48	Inverted-variate	0.28438	0.32693	0.33336
Hanwha	Full	H48	Joint-patch	0.36652	0.34526	0.33407
Hanwha	Full	H48	Depthwise TCN	0.33142	0.33673	0.34221
Hanwha	Daylight	H48	Discrete recurrent	0.52195	0.46421	0.62989
Hanwha	Daylight	H48	Inverted-variate	0.47541	0.54087	0.52920
Hanwha	Daylight	H48	Joint-patch	0.59312	0.52110	0.52779
Hanwha	Daylight	H48	Depthwise TCN	0.52569	0.53522	0.53927
Hanwha	Full	H96	Discrete recurrent	0.40774	0.40206	0.45383
Hanwha	Full	H96	Inverted-variate	0.49253	0.52614	0.55511
Hanwha	Full	H96	Joint-patch	0.52264	0.46726	0.46749
Hanwha	Full	H96	Depthwise TCN	0.47434	0.48192	0.49329
Hanwha	Daylight	H96	Discrete recurrent	0.81616	0.83369	0.93289
Hanwha	Daylight	H96	Inverted-variate	0.87000	0.91131	0.93783
Hanwha	Daylight	H96	Joint-patch	0.92079	0.77157	0.79984
Hanwha	Daylight	H96	Depthwise TCN	0.81254	0.81988	0.83932
Hanwha	Full	H144	Discrete recurrent	0.40071	0.38910	0.42640
Hanwha	Full	H144	Inverted-variate	0.48388	0.52199	0.53621
Hanwha	Full	H144	Joint-patch	0.47310	0.44847	0.44631
Hanwha	Full	H144	Depthwise TCN	0.44980	0.47854	0.47476
Hanwha	Daylight	H144	Discrete recurrent	0.81786	0.82031	0.89037
Hanwha	Daylight	H144	Inverted-variate	0.84028	0.88713	0.89231
Hanwha	Daylight	H144	Joint-patch	0.82794	0.73477	0.76446
Hanwha	Daylight	H144	Depthwise TCN	0.74890	0.81601	0.79182
Qcells	Full	H12	Discrete recurrent	0.26715	0.44107	0.92759
Qcells	Full	H12	Inverted-variate	0.17351	0.23238	0.20505
Qcells	Full	H12	Joint-patch	0.38756	0.38343	0.33033
Qcells	Full	H12	Depthwise TCN	0.33873	0.18634	0.23193
Qcells	Daylight	H12	Discrete recurrent	0.51963	0.87320	1.85785
Qcells	Daylight	H12	Inverted-variate	0.28354	0.34031	0.33801
Qcells	Daylight	H12	Joint-patch	0.59609	0.67195	0.56727
Qcells	Daylight	H12	Depthwise TCN	0.62824	0.31843	0.44563
Qcells	Full	H48	Discrete recurrent	0.38057	0.54036	1.10494
Qcells	Full	H48	Inverted-variate	0.42459	0.46561	0.44915
Qcells	Full	H48	Joint-patch	0.60764	0.63207	0.60534
Qcells	Full	H48	Depthwise TCN	0.70893	0.59737	0.57099
Qcells	Daylight	H48	Discrete recurrent	0.75199	1.05716	2.25583
Qcells	Daylight	H48	Inverted-variate	0.77200	0.81193	0.79177
Qcells	Daylight	H48	Joint-patch	1.03945	1.18366	1.13374
Qcells	Daylight	H48	Depthwise TCN	1.38908	1.15015	1.12203
Qcells	Full	H96	Discrete recurrent	0.54199	0.65278	0.90353
Qcells	Full	H96	Inverted-variate	0.66567	0.68290	0.66126
Qcells	Full	H96	Joint-patch	0.74340	0.75035	0.71193
Qcells	Full	H96	Depthwise TCN	0.78189	0.82302	0.73178
Qcells	Daylight	H96	Discrete recurrent	1.11654	1.31308	1.84620
Qcells	Daylight	H96	Inverted-variate	1.24135	1.24596	1.19846
Qcells	Daylight	H96	Joint-patch	1.33358	1.41995	1.33442

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Daylight	H96	Depthwise TCN	1.52256	1.61576	1.41140
Qcells	Full	H144	Discrete recurrent	0.49640	0.51181	0.48935
Qcells	Full	H144	Inverted-variate	0.52191	0.56507	0.51260
Qcells	Full	H144	Joint-patch	0.56861	0.53326	0.50553
Qcells	Full	H144	Depthwise TCN	0.49955	0.47670	0.47809
Qcells	Daylight	H144	Discrete recurrent	1.06417	1.04785	0.97894
Qcells	Daylight	H144	Inverted-variate	0.94392	0.98971	0.90037
Qcells	Daylight	H144	Joint-patch	0.97849	0.97913	0.92040
Qcells	Daylight	H144	Depthwise TCN	0.93775	0.88304	0.86744

Table S4: Primary per-seed R2 values.

Alt text: Per-seed coefficients of determination show horizon- and scope-dependent explained variance, including negative values where forecasts underperform the target mean. Values are descriptive because adjacent forecast origins overlap.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.99012	0.98966	0.99051
Sanyo	Full	H12	Inverted-variate	0.98379	0.98246	0.98810
Sanyo	Full	H12	Joint-patch	0.98017	0.98580	0.98890
Sanyo	Full	H12	Depthwise TCN	0.98743	0.99290	0.98508
Sanyo	Daylight	H12	Discrete recurrent	0.96350	0.96081	0.96469
Sanyo	Daylight	H12	Inverted-variate	0.95791	0.95588	0.96020
Sanyo	Daylight	H12	Joint-patch	0.94043	0.95655	0.96809
Sanyo	Daylight	H12	Depthwise TCN	0.95965	0.97592	0.95721
Sanyo	Full	H48	Discrete recurrent	0.90861	0.90941	0.90303
Sanyo	Full	H48	Inverted-variate	0.89887	0.91521	0.91574
Sanyo	Full	H48	Joint-patch	0.90236	0.92567	0.92301
Sanyo	Full	H48	Depthwise TCN	0.89986	0.90917	0.91200
Sanyo	Daylight	H48	Discrete recurrent	0.59798	0.60072	0.57216
Sanyo	Daylight	H48	Inverted-variate	0.59742	0.70551	0.67183
Sanyo	Daylight	H48	Joint-patch	0.59573	0.71394	0.68769
Sanyo	Daylight	H48	Depthwise TCN	0.60426	0.63806	0.66179
Sanyo	Full	H96	Discrete recurrent	0.81383	0.81722	0.78906
Sanyo	Full	H96	Inverted-variate	0.83419	0.85241	0.85188
Sanyo	Full	H96	Joint-patch	0.79010	0.83714	0.85173
Sanyo	Full	H96	Depthwise TCN	0.79050	0.83409	0.82312
Sanyo	Daylight	H96	Discrete recurrent	0.22692	0.24393	0.12087
Sanyo	Daylight	H96	Inverted-variate	0.39505	0.49534	0.46819
Sanyo	Daylight	H96	Joint-patch	0.16703	0.38723	0.43986
Sanyo	Daylight	H96	Depthwise TCN	0.20727	0.38910	0.34507
Sanyo	Full	H144	Discrete recurrent	0.81869	0.83965	0.80974
Sanyo	Full	H144	Inverted-variate	0.85662	0.86284	0.85625
Sanyo	Full	H144	Joint-patch	0.83160	0.85618	0.86775
Sanyo	Full	H144	Depthwise TCN	0.82702	0.85094	0.85096
Sanyo	Daylight	H144	Discrete recurrent	0.30739	0.39245	0.27621
Sanyo	Daylight	H144	Inverted-variate	0.53928	0.57615	0.53471
Sanyo	Daylight	H144	Joint-patch	0.39957	0.50855	0.54701
Sanyo	Daylight	H144	Depthwise TCN	0.41323	0.50541	0.50270

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Hanwha	Full	H12	Discrete recurrent	0.98994	0.98809	0.98471
Hanwha	Full	H12	Inverted-variate	0.98989	0.98780	0.98778
Hanwha	Full	H12	Joint-patch	0.98303	0.97656	0.98538
Hanwha	Full	H12	Depthwise TCN	0.99128	0.99240	0.99127
Hanwha	Daylight	H12	Discrete recurrent	0.96479	0.95696	0.94580
Hanwha	Daylight	H12	Inverted-variate	0.96898	0.96178	0.96210
Hanwha	Daylight	H12	Joint-patch	0.95397	0.95383	0.96269
Hanwha	Daylight	H12	Depthwise TCN	0.97447	0.97724	0.97347
Hanwha	Full	H48	Discrete recurrent	0.91295	0.91474	0.89469
Hanwha	Full	H48	Inverted-variate	0.92245	0.91005	0.91099
Hanwha	Full	H48	Joint-patch	0.90099	0.91483	0.91708
Hanwha	Full	H48	Depthwise TCN	0.91358	0.91121	0.90760
Hanwha	Daylight	H48	Discrete recurrent	0.65097	0.65314	0.57294
Hanwha	Daylight	H48	Inverted-variate	0.70142	0.65611	0.66810
Hanwha	Daylight	H48	Joint-patch	0.62249	0.68809	0.69251
Hanwha	Daylight	H48	Depthwise TCN	0.67831	0.67232	0.65654
Hanwha	Full	H96	Discrete recurrent	0.82583	0.80303	0.78172
Hanwha	Full	H96	Inverted-variate	0.81133	0.80494	0.78898
Hanwha	Full	H96	Joint-patch	0.79410	0.84324	0.83711
Hanwha	Full	H96	Depthwise TCN	0.83056	0.82993	0.82255
Hanwha	Daylight	H96	Discrete recurrent	0.33672	0.24166	0.15955
Hanwha	Daylight	H96	Inverted-variate	0.31390	0.30246	0.25794
Hanwha	Daylight	H96	Joint-patch	0.24331	0.44578	0.42006
Hanwha	Daylight	H96	Depthwise TCN	0.40080	0.41041	0.37877
Hanwha	Full	H144	Discrete recurrent	0.84051	0.83045	0.81835
Hanwha	Full	H144	Inverted-variate	0.82978	0.82222	0.81141
Hanwha	Full	H144	Joint-patch	0.82575	0.86041	0.85248
Hanwha	Full	H144	Depthwise TCN	0.85489	0.84483	0.84345
Hanwha	Daylight	H144	Discrete recurrent	0.43177	0.39155	0.34756
Hanwha	Daylight	H144	Inverted-variate	0.43445	0.42068	0.39739
Hanwha	Daylight	H144	Joint-patch	0.41750	0.54985	0.51736
Hanwha	Daylight	H144	Depthwise TCN	0.54036	0.50451	0.50586
Qcells	Full	H12	Discrete recurrent	0.94988	0.86178	0.35216
Qcells	Full	H12	Inverted-variate	0.97671	0.96878	0.97199
Qcells	Full	H12	Joint-patch	0.92817	0.91549	0.93806
Qcells	Full	H12	Depthwise TCN	0.92923	0.97608	0.95563
Qcells	Daylight	H12	Discrete recurrent	0.81396	0.48674	-1.41663
Qcells	Daylight	H12	Inverted-variate	0.91752	0.89828	0.90184
Qcells	Daylight	H12	Joint-patch	0.76075	0.69767	0.78069
Qcells	Daylight	H12	Depthwise TCN	0.74107	0.91487	0.83603
Qcells	Full	H48	Discrete recurrent	0.85828	0.78869	0.20558
Qcells	Full	H48	Inverted-variate	0.85194	0.84060	0.84465
Qcells	Full	H48	Joint-patch	0.79198	0.74781	0.76963
Qcells	Full	H48	Depthwise TCN	0.65567	0.74045	0.74340
Qcells	Daylight	H48	Discrete recurrent	0.37554	0.07240	-2.51653
Qcells	Daylight	H48	Inverted-variate	0.36053	0.31768	0.33533
Qcells	Daylight	H48	Joint-patch	0.11900	-0.09771	-0.00327
Qcells	Daylight	H48	Depthwise TCN	-0.51269	-0.13879	-0.12748
Qcells	Full	H96	Discrete recurrent	0.72851	0.66714	0.39676

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Full	H96	Inverted-variate	0.67189	0.67404	0.69231
Qcells	Full	H96	Joint-patch	0.65344	0.62281	0.66015
Qcells	Full	H96	Depthwise TCN	0.54001	0.48761	0.59396
Qcells	Daylight	H96	Discrete recurrent	-0.08402	-0.31618	-1.40133
Qcells	Daylight	H96	Inverted-variate	-0.27312	-0.25659	-0.17520
Qcells	Daylight	H96	Joint-patch	-0.33529	-0.47135	-0.32141
Qcells	Daylight	H96	Depthwise TCN	-0.81087	-1.02099	-0.59047
Qcells	Full	H144	Discrete recurrent	0.78078	0.77078	0.78948
Qcells	Full	H144	Inverted-variate	0.80940	0.79931	0.82296
Qcells	Full	H144	Joint-patch	0.79847	0.80546	0.81892
Qcells	Full	H144	Depthwise TCN	0.82984	0.83272	0.84063
Qcells	Daylight	H144	Discrete recurrent	0.22854	0.20807	0.27726
Qcells	Daylight	H144	Inverted-variate	0.37117	0.34867	0.43114
Qcells	Daylight	H144	Joint-patch	0.34174	0.35260	0.39928
Qcells	Daylight	H144	Depthwise TCN	0.43312	0.44472	0.47782

Table S5: Primary per-seed Bias values.

Alt text: Per-seed mean errors identify underprediction and overprediction patterns across the three arrays, two scopes, and four forecast horizons. Positive values indicate overprediction, and negative values indicate underprediction.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	-0.01485	0.02522	-0.01926
Sanyo	Full	H12	Inverted-variate	0.06001	0.08298	-0.00450
Sanyo	Full	H12	Joint-patch	0.05334	0.07973	0.05282
Sanyo	Full	H12	Depthwise TCN	-0.06692	0.03892	0.16604
Sanyo	Daylight	H12	Discrete recurrent	-0.09490	0.04305	-0.06442
Sanyo	Daylight	H12	Inverted-variate	-0.06615	0.02881	-0.02995
Sanyo	Daylight	H12	Joint-patch	0.08527	0.10689	0.06451
Sanyo	Daylight	H12	Depthwise TCN	-0.22977	0.06247	0.18508
Sanyo	Full	H48	Discrete recurrent	-0.19674	-0.14171	-0.18156
Sanyo	Full	H48	Inverted-variate	-0.15590	0.00905	-0.13983
Sanyo	Full	H48	Joint-patch	-0.19055	-0.04334	-0.12676
Sanyo	Full	H48	Depthwise TCN	-0.19644	-0.14263	-0.07699
Sanyo	Daylight	H48	Discrete recurrent	-0.49498	-0.33281	-0.45605
Sanyo	Daylight	H48	Inverted-variate	-0.54239	-0.31128	-0.46493
Sanyo	Daylight	H48	Joint-patch	-0.44354	-0.25136	-0.35105
Sanyo	Daylight	H48	Depthwise TCN	-0.58612	-0.43602	-0.37384
Sanyo	Full	H96	Discrete recurrent	-0.33184	-0.30250	-0.35482
Sanyo	Full	H96	Inverted-variate	-0.21373	-0.09706	-0.20703
Sanyo	Full	H96	Joint-patch	-0.36865	-0.22148	-0.22546
Sanyo	Full	H96	Depthwise TCN	-0.35401	-0.19358	-0.20143
Sanyo	Daylight	H96	Discrete recurrent	-0.80950	-0.71352	-0.84974
Sanyo	Daylight	H96	Inverted-variate	-0.66186	-0.54659	-0.61488
Sanyo	Daylight	H96	Joint-patch	-0.84300	-0.59778	-0.54960
Sanyo	Daylight	H96	Depthwise TCN	-0.90937	-0.60478	-0.67695
Sanyo	Full	H144	Discrete recurrent	-0.30150	-0.24728	-0.28443
Sanyo	Full	H144	Inverted-variate	-0.12415	-0.07899	-0.19460

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H144	Joint-patch	-0.22895	-0.16462	-0.18296
Sanyo	Full	H144	Depthwise TCN	-0.25089	-0.17817	-0.16368
Sanyo	Daylight	H144	Discrete recurrent	-0.76490	-0.61458	-0.72355
Sanyo	Daylight	H144	Inverted-variate	-0.45553	-0.45708	-0.55347
Sanyo	Daylight	H144	Joint-patch	-0.57689	-0.49772	-0.44642
Sanyo	Daylight	H144	Depthwise TCN	-0.70901	-0.59856	-0.57486
Hanwha	Full	H12	Discrete recurrent	-0.07223	-0.06327	-0.12246
Hanwha	Full	H12	Inverted-variate	-0.00845	-0.01508	-0.01574
Hanwha	Full	H12	Joint-patch	0.09252	-0.19886	-0.05599
Hanwha	Full	H12	Depthwise TCN	-0.02420	0.01298	0.03659
Hanwha	Daylight	H12	Discrete recurrent	-0.13923	-0.12002	-0.21837
Hanwha	Daylight	H12	Inverted-variate	0.00506	-0.08452	-0.07401
Hanwha	Daylight	H12	Joint-patch	0.06763	-0.15859	0.00038
Hanwha	Daylight	H12	Depthwise TCN	-0.03043	-0.00929	0.05944
Hanwha	Full	H48	Discrete recurrent	-0.21430	-0.17656	-0.27482
Hanwha	Full	H48	Inverted-variate	-0.13472	-0.18278	-0.14005
Hanwha	Full	H48	Joint-patch	-0.15171	-0.23340	-0.19411
Hanwha	Full	H48	Depthwise TCN	-0.19526	-0.16190	-0.18508
Hanwha	Daylight	H48	Discrete recurrent	-0.46980	-0.38290	-0.56927
Hanwha	Daylight	H48	Inverted-variate	-0.35116	-0.44743	-0.40447
Hanwha	Daylight	H48	Joint-patch	-0.45830	-0.40349	-0.37726
Hanwha	Daylight	H48	Depthwise TCN	-0.41941	-0.45174	-0.41057
Hanwha	Full	H96	Discrete recurrent	-0.32869	-0.33349	-0.39879
Hanwha	Full	H96	Inverted-variate	-0.32324	-0.34001	-0.34498
Hanwha	Full	H96	Joint-patch	-0.33245	-0.29819	-0.28419
Hanwha	Full	H96	Depthwise TCN	-0.26653	-0.22095	-0.26397
Hanwha	Daylight	H96	Discrete recurrent	-0.72991	-0.74335	-0.86099
Hanwha	Daylight	H96	Inverted-variate	-0.72901	-0.76600	-0.78191
Hanwha	Daylight	H96	Joint-patch	-0.77479	-0.55500	-0.59110
Hanwha	Daylight	H96	Depthwise TCN	-0.62178	-0.63182	-0.64313
Hanwha	Full	H144	Discrete recurrent	-0.27785	-0.26207	-0.31324
Hanwha	Full	H144	Inverted-variate	-0.27110	-0.30315	-0.28482
Hanwha	Full	H144	Joint-patch	-0.20469	-0.24152	-0.21510
Hanwha	Full	H144	Depthwise TCN	-0.17934	-0.19233	-0.18866
Hanwha	Daylight	H144	Discrete recurrent	-0.62704	-0.60357	-0.69016
Hanwha	Daylight	H144	Inverted-variate	-0.60275	-0.65849	-0.64942
Hanwha	Daylight	H144	Joint-patch	-0.52855	-0.43327	-0.48145
Hanwha	Daylight	H144	Depthwise TCN	-0.46343	-0.56530	-0.50861
Qcells	Full	H12	Discrete recurrent	-0.13648	-0.30177	-0.82254
Qcells	Full	H12	Inverted-variate	-0.01945	-0.08940	-0.05842
Qcells	Full	H12	Joint-patch	-0.04125	-0.14744	-0.20265
Qcells	Full	H12	Depthwise TCN	-0.19140	-0.08991	0.01064
Qcells	Daylight	H12	Discrete recurrent	-0.30502	-0.68714	-1.67725
Qcells	Daylight	H12	Inverted-variate	-0.01510	-0.10341	-0.09454
Qcells	Daylight	H12	Joint-patch	-0.30270	-0.42471	-0.34917
Qcells	Daylight	H12	Depthwise TCN	-0.46033	-0.15339	0.02249
Qcells	Full	H48	Discrete recurrent	-0.26981	-0.42263	-1.02588
Qcells	Full	H48	Inverted-variate	-0.28195	-0.33935	-0.29151
Qcells	Full	H48	Joint-patch	-0.32374	-0.46845	-0.52214

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Full	H48	Depthwise TCN	-0.59271	-0.51875	-0.42368
Qcells	Daylight	H48	Discrete recurrent	-0.60787	-0.97543	-2.18057
Qcells	Daylight	H48	Inverted-variate	-0.59994	-0.65309	-0.61191
Qcells	Daylight	H48	Joint-patch	-0.92710	-1.09891	-1.06507
Qcells	Daylight	H48	Depthwise TCN	-1.33813	-1.10355	-0.93068
Qcells	Full	H96	Discrete recurrent	-0.41366	-0.49742	-0.76850
Qcells	Full	H96	Inverted-variate	-0.44038	-0.45492	-0.37560
Qcells	Full	H96	Joint-patch	-0.45584	-0.54116	-0.55976
Qcells	Full	H96	Depthwise TCN	-0.59562	-0.65654	-0.54134
Qcells	Daylight	H96	Discrete recurrent	-0.95209	-1.17593	-1.69645
Qcells	Daylight	H96	Inverted-variate	-0.96993	-0.98873	-0.91253
Qcells	Daylight	H96	Joint-patch	-1.15251	-1.26438	-1.17279
Qcells	Daylight	H96	Depthwise TCN	-1.38535	-1.48752	-1.19143
Qcells	Full	H144	Discrete recurrent	-0.35576	-0.27276	-0.26605
Qcells	Full	H144	Inverted-variate	-0.28983	-0.33920	-0.20222
Qcells	Full	H144	Joint-patch	-0.25484	-0.28206	-0.29352
Qcells	Full	H144	Depthwise TCN	-0.25781	-0.22607	-0.25775
Qcells	Daylight	H144	Discrete recurrent	-0.85934	-0.71506	-0.62348
Qcells	Daylight	H144	Inverted-variate	-0.63709	-0.71159	-0.54233
Qcells	Daylight	H144	Joint-patch	-0.69371	-0.70131	-0.62555
Qcells	Daylight	H144	Depthwise TCN	-0.68415	-0.55354	-0.55718

Table S6: Primary per-seed range_nRMSE values.

Alt text: Per-seed Train-range normalized RMSE values permit within-array comparison of error growth while avoiding unsupported AC-capacity normalization. The denominator is fitted only on Train and differs by array.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.03417	0.03495	0.03348
Sanyo	Full	H12	Inverted-variate	0.04377	0.04553	0.03750
Sanyo	Full	H12	Joint-patch	0.04841	0.04097	0.03622
Sanyo	Full	H12	Depthwise TCN	0.03855	0.02897	0.04199
Sanyo	Daylight	H12	Discrete recurrent	0.04886	0.05064	0.04806
Sanyo	Daylight	H12	Inverted-variate	0.05248	0.05373	0.05103
Sanyo	Daylight	H12	Joint-patch	0.06242	0.05331	0.04569
Sanyo	Daylight	H12	Depthwise TCN	0.05138	0.03969	0.05291
Sanyo	Full	H48	Discrete recurrent	0.10678	0.10631	0.10999
Sanyo	Full	H48	Inverted-variate	0.11232	0.10285	0.10253
Sanyo	Full	H48	Joint-patch	0.11037	0.09630	0.09800
Sanyo	Full	H48	Depthwise TCN	0.11177	0.10645	0.10477
Sanyo	Daylight	H48	Discrete recurrent	0.15678	0.15624	0.16174
Sanyo	Daylight	H48	Inverted-variate	0.15689	0.13418	0.14165
Sanyo	Daylight	H48	Joint-patch	0.15722	0.13225	0.13818
Sanyo	Daylight	H48	Depthwise TCN	0.15555	0.14876	0.14380
Sanyo	Full	H96	Discrete recurrent	0.14995	0.14859	0.15962
Sanyo	Full	H96	Inverted-variate	0.14152	0.13352	0.13375
Sanyo	Full	H96	Joint-patch	0.15922	0.14025	0.13383
Sanyo	Full	H96	Depthwise TCN	0.15907	0.14156	0.14616
Sanyo	Daylight	H96	Discrete recurrent	0.22262	0.22016	0.23740

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Daylight	H96	Inverted-variate	0.19693	0.17987	0.18464
Sanyo	Daylight	H96	Joint-patch	0.23108	0.19820	0.18950
Sanyo	Daylight	H96	Depthwise TCN	0.22543	0.19790	0.20490
Sanyo	Full	H144	Discrete recurrent	0.14497	0.13634	0.14850
Sanyo	Full	H144	Inverted-variate	0.12892	0.12609	0.12908
Sanyo	Full	H144	Joint-patch	0.13971	0.12911	0.12381
Sanyo	Full	H144	Depthwise TCN	0.14160	0.13145	0.13144
Sanyo	Daylight	H144	Discrete recurrent	0.21658	0.20284	0.22140
Sanyo	Daylight	H144	Inverted-variate	0.17664	0.16942	0.17751
Sanyo	Daylight	H144	Joint-patch	0.20165	0.18243	0.17515
Sanyo	Daylight	H144	Depthwise TCN	0.19934	0.18301	0.18352
Hanwha	Full	H12	Discrete recurrent	0.03308	0.03600	0.04078
Hanwha	Full	H12	Inverted-variate	0.03316	0.03643	0.03646
Hanwha	Full	H12	Joint-patch	0.04297	0.05050	0.03988
Hanwha	Full	H12	Depthwise TCN	0.03079	0.02876	0.03081
Hanwha	Daylight	H12	Discrete recurrent	0.04703	0.05200	0.05835
Hanwha	Daylight	H12	Inverted-variate	0.04414	0.04900	0.04879
Hanwha	Daylight	H12	Joint-patch	0.05377	0.05385	0.04841
Hanwha	Daylight	H12	Depthwise TCN	0.04005	0.03781	0.04082
Hanwha	Full	H48	Discrete recurrent	0.09950	0.09847	0.10944
Hanwha	Full	H48	Inverted-variate	0.09391	0.10114	0.10061
Hanwha	Full	H48	Joint-patch	0.10611	0.09842	0.09711
Hanwha	Full	H48	Depthwise TCN	0.09914	0.10049	0.10251
Hanwha	Daylight	H48	Discrete recurrent	0.14534	0.14489	0.16077
Hanwha	Daylight	H48	Inverted-variate	0.13443	0.14427	0.14173
Hanwha	Daylight	H48	Joint-patch	0.15115	0.13740	0.13642
Hanwha	Daylight	H48	Depthwise TCN	0.13953	0.14083	0.14418
Hanwha	Full	H96	Discrete recurrent	0.13866	0.14746	0.15523
Hanwha	Full	H96	Inverted-variate	0.14432	0.14674	0.15263
Hanwha	Full	H96	Joint-patch	0.15076	0.13155	0.13410
Hanwha	Full	H96	Depthwise TCN	0.13677	0.13702	0.13996
Hanwha	Daylight	H96	Discrete recurrent	0.20444	0.21860	0.23013
Hanwha	Daylight	H96	Inverted-variate	0.20792	0.20965	0.21624
Hanwha	Daylight	H96	Joint-patch	0.21836	0.18687	0.19116
Hanwha	Daylight	H96	Depthwise TCN	0.19431	0.19275	0.19785
Hanwha	Full	H144	Discrete recurrent	0.13050	0.13455	0.13927
Hanwha	Full	H144	Inverted-variate	0.13482	0.13778	0.14191
Hanwha	Full	H144	Joint-patch	0.13640	0.12208	0.12550
Hanwha	Full	H144	Depthwise TCN	0.12448	0.12872	0.12929
Hanwha	Daylight	H144	Discrete recurrent	0.19335	0.20007	0.20718
Hanwha	Daylight	H144	Inverted-variate	0.19289	0.19523	0.19911
Hanwha	Daylight	H144	Joint-patch	0.19576	0.17209	0.17819
Hanwha	Daylight	H144	Depthwise TCN	0.17390	0.18055	0.18030
Qcells	Full	H12	Discrete recurrent	0.07332	0.12176	0.26361
Qcells	Full	H12	Inverted-variate	0.04999	0.05787	0.05482
Qcells	Full	H12	Joint-patch	0.08778	0.09521	0.08151
Qcells	Full	H12	Depthwise TCN	0.08713	0.05065	0.06899
Qcells	Daylight	H12	Discrete recurrent	0.10642	0.17677	0.38357
Qcells	Daylight	H12	Inverted-variate	0.07086	0.07869	0.07731

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Daylight	H12	Joint-patch	0.12069	0.13567	0.11555
Qcells	Daylight	H12	Depthwise TCN	0.12556	0.07199	0.09991
Qcells	Full	H48	Discrete recurrent	0.12822	0.15657	0.30358
Qcells	Full	H48	Inverted-variate	0.13106	0.13598	0.13425
Qcells	Full	H48	Joint-patch	0.15535	0.17104	0.16348
Qcells	Full	H48	Depthwise TCN	0.19986	0.17352	0.17253
Qcells	Daylight	H48	Discrete recurrent	0.18663	0.22746	0.44287
Qcells	Daylight	H48	Inverted-variate	0.18886	0.19508	0.19254
Qcells	Daylight	H48	Joint-patch	0.22167	0.24744	0.23655
Qcells	Daylight	H48	Depthwise TCN	0.29047	0.25202	0.25077
Qcells	Full	H96	Discrete recurrent	0.17328	0.19187	0.25830
Qcells	Full	H96	Inverted-variate	0.19050	0.18988	0.18448
Qcells	Full	H96	Joint-patch	0.19578	0.20425	0.19388
Qcells	Full	H96	Depthwise TCN	0.22556	0.23806	0.21192
Qcells	Daylight	H96	Discrete recurrent	0.25569	0.28174	0.38055
Qcells	Daylight	H96	Inverted-variate	0.27709	0.27529	0.26622
Qcells	Daylight	H96	Joint-patch	0.28378	0.29788	0.28230
Qcells	Daylight	H96	Depthwise TCN	0.33047	0.34912	0.30971
Qcells	Full	H144	Discrete recurrent	0.15035	0.15374	0.14734
Qcells	Full	H144	Inverted-variate	0.14019	0.14386	0.13512
Qcells	Full	H144	Joint-patch	0.14416	0.14164	0.13665
Qcells	Full	H144	Depthwise TCN	0.13246	0.13134	0.12819
Qcells	Daylight	H144	Discrete recurrent	0.22649	0.22948	0.21922
Qcells	Daylight	H144	Inverted-variate	0.20448	0.20811	0.19449
Qcells	Daylight	H144	Joint-patch	0.20921	0.20748	0.19986
Qcells	Daylight	H144	Depthwise TCN	0.19415	0.19215	0.18634

Table S7: Primary per-seed RMSE_skill values.

Alt text: Per-seed RMSE skill relative to Last-value Persistence quantifies improvement beyond local continuity; positive values favor the neural implementation. Every value uses the same origins, labels, and masks as its reference.

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	H12	Discrete recurrent	0.55381	0.54361	0.56280
Sanyo	Full	H12	Inverted-variate	0.42848	0.40540	0.51032
Sanyo	Full	H12	Joint-patch	0.36784	0.46504	0.52699
Sanyo	Full	H12	Depthwise TCN	0.49666	0.62169	0.45164
Sanyo	Daylight	H12	Discrete recurrent	0.55629	0.54019	0.56358
Sanyo	Daylight	H12	Inverted-variate	0.52348	0.51213	0.53664
Sanyo	Daylight	H12	Joint-patch	0.43314	0.51588	0.58512
Sanyo	Daylight	H12	Depthwise TCN	0.53347	0.63962	0.51956
Sanyo	Full	H48	Discrete recurrent	0.55058	0.55255	0.53706
Sanyo	Full	H48	Inverted-variate	0.52724	0.56711	0.56847
Sanyo	Full	H48	Joint-patch	0.53546	0.59469	0.58751
Sanyo	Full	H48	Depthwise TCN	0.52955	0.55195	0.55900
Sanyo	Daylight	H48	Discrete recurrent	0.53089	0.53249	0.51606
Sanyo	Daylight	H48	Inverted-variate	0.53056	0.59850	0.57616
Sanyo	Daylight	H48	Joint-patch	0.52958	0.60429	0.58653

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Sanyo	Daylight	H48	Depthwise TCN	0.53457	0.55489	0.56973
Sanyo	Full	H96	Discrete recurrent	0.60553	0.60913	0.58010
Sanyo	Full	H96	Inverted-variate	0.62772	0.64877	0.64814
Sanyo	Full	H96	Joint-patch	0.58114	0.63104	0.64795
Sanyo	Full	H96	Depthwise TCN	0.58154	0.62760	0.61550
Sanyo	Daylight	H96	Discrete recurrent	0.57233	0.57706	0.54394
Sanyo	Daylight	H96	Inverted-variate	0.62169	0.65447	0.64529
Sanyo	Daylight	H96	Joint-patch	0.55608	0.61925	0.63597
Sanyo	Daylight	H96	Depthwise TCN	0.56693	0.61983	0.60637
Sanyo	Full	H144	Discrete recurrent	0.67824	0.69741	0.67040
Sanyo	Full	H144	Inverted-variate	0.71387	0.72014	0.71350
Sanyo	Full	H144	Joint-patch	0.68991	0.71343	0.72520
Sanyo	Full	H144	Depthwise TCN	0.68572	0.70826	0.70828
Sanyo	Daylight	H144	Discrete recurrent	0.64239	0.66507	0.63443
Sanyo	Daylight	H144	Inverted-variate	0.70834	0.72025	0.70689
Sanyo	Daylight	H144	Joint-patch	0.66704	0.69876	0.71079
Sanyo	Daylight	H144	Depthwise TCN	0.67084	0.69781	0.69698
Hanwha	Full	H12	Discrete recurrent	0.55574	0.51655	0.45229
Hanwha	Full	H12	Inverted-variate	0.55474	0.51080	0.51031
Hanwha	Full	H12	Joint-patch	0.42297	0.32186	0.46437
Hanwha	Full	H12	Depthwise TCN	0.58645	0.61382	0.58622
Hanwha	Daylight	H12	Discrete recurrent	0.56394	0.51785	0.45895
Hanwha	Daylight	H12	Inverted-variate	0.59070	0.54567	0.54755
Hanwha	Daylight	H12	Joint-patch	0.50142	0.50064	0.55112
Hanwha	Daylight	H12	Depthwise TCN	0.62865	0.64938	0.62148
Hanwha	Full	H48	Discrete recurrent	0.57112	0.57557	0.52829
Hanwha	Full	H48	Inverted-variate	0.59521	0.56405	0.56633
Hanwha	Full	H48	Joint-patch	0.54262	0.57578	0.58142
Hanwha	Full	H48	Depthwise TCN	0.57268	0.56686	0.55815
Hanwha	Daylight	H48	Discrete recurrent	0.54906	0.55046	0.50119
Hanwha	Daylight	H48	Inverted-variate	0.58292	0.55239	0.56026
Hanwha	Daylight	H48	Joint-patch	0.53102	0.57371	0.57674
Hanwha	Daylight	H48	Depthwise TCN	0.56708	0.56307	0.55267
Hanwha	Full	H96	Discrete recurrent	0.62830	0.60473	0.58390
Hanwha	Full	H96	Inverted-variate	0.61315	0.60664	0.59087
Hanwha	Full	H96	Joint-patch	0.59587	0.64737	0.64055
Hanwha	Full	H96	Depthwise TCN	0.63338	0.63270	0.62482
Hanwha	Daylight	H96	Discrete recurrent	0.58419	0.55539	0.53194
Hanwha	Daylight	H96	Inverted-variate	0.57710	0.57359	0.56019
Hanwha	Daylight	H96	Joint-patch	0.55587	0.61991	0.61119
Hanwha	Daylight	H96	Depthwise TCN	0.60478	0.60797	0.59758
Hanwha	Full	H144	Discrete recurrent	0.70299	0.69377	0.68303
Hanwha	Full	H144	Inverted-variate	0.69316	0.68642	0.67703
Hanwha	Full	H144	Joint-patch	0.68956	0.72214	0.71436
Hanwha	Full	H144	Depthwise TCN	0.71670	0.70704	0.70574
Hanwha	Daylight	H144	Discrete recurrent	0.65561	0.64363	0.63098
Hanwha	Daylight	H144	Inverted-variate	0.65643	0.65227	0.64535
Hanwha	Daylight	H144	Joint-patch	0.65132	0.69348	0.68261
Hanwha	Daylight	H144	Depthwise TCN	0.69026	0.67841	0.67885

Array	Scope	Horizon	Model	Seed 42	Seed 43	Seed 44
Qcells	Full	H12	Discrete recurrent	0.05980	-0.56142	-2.38038
Qcells	Full	H12	Inverted-variate	0.35900	0.25792	0.29709
Qcells	Full	H12	Joint-patch	-0.12558	-0.22089	-0.04527
Qcells	Full	H12	Depthwise TCN	-0.11728	0.35050	0.11532
Qcells	Daylight	H12	Discrete recurrent	0.05756	-0.56540	-2.39674
Qcells	Daylight	H12	Inverted-variate	0.37248	0.30313	0.31542
Qcells	Daylight	H12	Joint-patch	-0.06876	-0.20143	-0.02325
Qcells	Daylight	H12	Depthwise TCN	-0.11186	0.36248	0.11521
Qcells	Full	H48	Discrete recurrent	0.40800	0.27713	-0.40160
Qcells	Full	H48	Inverted-variate	0.39491	0.37217	0.38020
Qcells	Full	H48	Joint-patch	0.28278	0.21030	0.24524
Qcells	Full	H48	Depthwise TCN	0.07725	0.19886	0.20342
Qcells	Daylight	H48	Discrete recurrent	0.40872	0.27935	-0.40314
Qcells	Daylight	H48	Inverted-variate	0.40165	0.38193	0.38998
Qcells	Daylight	H48	Joint-patch	0.29769	0.21605	0.25054
Qcells	Daylight	H48	Depthwise TCN	0.07972	0.20152	0.20550
Qcells	Full	H96	Discrete recurrent	0.49082	0.43619	0.24100
Qcells	Full	H96	Inverted-variate	0.44023	0.44207	0.45793
Qcells	Full	H96	Joint-patch	0.42471	0.39982	0.43031
Qcells	Full	H96	Depthwise TCN	0.33721	0.30048	0.37729
Qcells	Daylight	H96	Discrete recurrent	0.49248	0.44077	0.24463
Qcells	Daylight	H96	Inverted-variate	0.44999	0.45357	0.47157
Qcells	Daylight	H96	Joint-patch	0.43672	0.40872	0.43966
Qcells	Daylight	H96	Depthwise TCN	0.34404	0.30703	0.38525
Qcells	Full	H144	Discrete recurrent	0.62459	0.61612	0.63211
Qcells	Full	H144	Inverted-variate	0.64995	0.64080	0.66263
Qcells	Full	H144	Joint-patch	0.64005	0.64635	0.65881
Qcells	Full	H144	Depthwise TCN	0.66925	0.67206	0.67992
Qcells	Daylight	H144	Discrete recurrent	0.62674	0.62182	0.63871
Qcells	Daylight	H144	Inverted-variate	0.66300	0.65703	0.67948
Qcells	Daylight	H144	Joint-patch	0.65521	0.65806	0.67062
Qcells	Daylight	H144	Depthwise TCN	0.68003	0.68333	0.69291

Table S8: Primary horizon-specific evaluation support.

Alt text: Forecast-origin and valid-target counts decrease with horizon and are shared by every method within each array-specific comparison. Full and daylight scopes report their target support separately.

Array	Scope	Horizon	Forecast origins	Valid target points
Sanyo	Full	H12	6,589	79,068
Sanyo	Daylight	H12	3,201	36,190
Sanyo	Full	H48	5,905	283,440
Sanyo	Daylight	H48	3,389	129,528
Sanyo	Full	H96	5,024	482,304
Sanyo	Daylight	H96	3,629	215,712
Sanyo	Full	H144	4,160	599,040
Sanyo	Daylight	H144	3,869	259,345
Hanwha	Full	H12	6,625	79,500
Hanwha	Daylight	H12	3,225	36,421

Array	Scope	Horizon	Forecast origins	Valid target points
Hanwha	Full	H48	6,049	290,352
Hanwha	Daylight	H48	3,522	132,649
Hanwha	Full	H96	5,281	506,976
Hanwha	Daylight	H96	3,881	228,420
Hanwha	Full	H144	4,521	651,024
Hanwha	Daylight	H144	4,225	287,368
Qcells	Full	H12	6,463	77,556
Qcells	Daylight	H12	3,173	36,504
Qcells	Full	H48	5,491	263,568
Qcells	Daylight	H48	3,071	123,447
Qcells	Full	H96	4,227	405,792
Qcells	Daylight	H96	2,927	184,959
Qcells	Full	H144	2,996	431,424
Qcells	Daylight	H144	2,800	187,950

Table S9: Complete-H144-origin sensitivity: Train-range nRMSE for every evaluated prefix. All horizons within a row use origins that possess a complete valid H144 target.

Alt text: Restricting every prefix to complete H144 origins preserves the main pattern of horizon-dependent error without changing the primary horizon-specific analysis. This analysis is secondary to the horizon-specific evaluation.

Array	Scope	Model	H12	H48	H96	H144
Sanyo	Full	Last-value	0.0491	0.1857	0.3539	0.4506
Sanyo	Full	Discrete recurrent	0.0230	0.0958	0.1458	0.1433
Sanyo	Full	Inverted-variate	0.0358	0.0948	0.1261	0.1280
Sanyo	Full	Joint-patch	0.0313	0.0907	0.1346	0.1309
Sanyo	Full	Depthwise TCN	0.0265	0.0950	0.1358	0.1348
Sanyo	Daylight	Last-value	0.1092	0.3385	0.5383	0.6056
Sanyo	Daylight	Discrete recurrent	0.0488	0.1929	0.2490	0.2136
Sanyo	Daylight	Inverted-variate	0.0525	0.1655	0.1921	0.1745
Sanyo	Daylight	Joint-patch	0.0492	0.1680	0.2175	0.1864
Sanyo	Daylight	Depthwise TCN	0.0405	0.1712	0.2137	0.1886
Hanwha	Full	Last-value	0.0591	0.2003	0.3546	0.4394
Hanwha	Full	Discrete recurrent	0.0309	0.0920	0.1381	0.1348
Hanwha	Full	Inverted-variate	0.0316	0.0907	0.1396	0.1382
Hanwha	Full	Joint-patch	0.0419	0.0937	0.1324	0.1280
Hanwha	Full	Depthwise TCN	0.0284	0.0936	0.1299	0.1275
Hanwha	Daylight	Last-value	0.1109	0.3252	0.4982	0.5614
Hanwha	Daylight	Discrete recurrent	0.0562	0.1631	0.2231	0.2002
Hanwha	Daylight	Inverted-variate	0.0509	0.1531	0.2153	0.1957
Hanwha	Daylight	Joint-patch	0.0539	0.1562	0.2052	0.1820
Hanwha	Daylight	Depthwise TCN	0.0453	0.1553	0.1974	0.1782
Qcells	Full	Last-value	0.0007	0.1037	0.2873	0.4005
Qcells	Full	Discrete recurrent	0.0138	0.0849	0.1527	0.1505
Qcells	Full	Inverted-variate	0.0207	0.0829	0.1411	0.1397
Qcells	Full	Joint-patch	0.0291	0.0849	0.1452	0.1408
Qcells	Full	Depthwise TCN	0.0129	0.0818	0.1334	0.1307

Array	Scope	Model	H12	H48	H96	H144
Qcells	Daylight	Last-value	0.0179	0.3209	0.5541	0.6068
Qcells	Daylight	Discrete recurrent	0.0219	0.2552	0.2898	0.2251
Qcells	Daylight	Inverted-variate	0.0315	0.2371	0.2574	0.2024
Qcells	Daylight	Joint-patch	0.0222	0.2382	0.2671	0.2055
Qcells	Daylight	Depthwise TCN	0.0201	0.2407	0.2456	0.1909

Table S10: Daily-matched RMSE (kW); the Daily-valid point mask is identical for all methods.

Alt text: On identical Daily-valid target points, Daily Persistence has lower RMSE than the post hoc neural envelope in 22 of 24 comparisons; the exceptions are Hanwha H12 full and daylight.

Array	Scope	Horizon	Daily	Last	Discrete	Inverted	Joint-patch	Depthwise
Sanyo	Full	H12	0.1752	0.4637	0.2072	0.2560	0.2535	0.2212
Sanyo	Daylight	H12	0.2589	0.6671	0.2980	0.3175	0.3260	0.2907
Sanyo	Full	H48	0.1595	1.4371	0.6527	0.6417	0.6153	0.6524
Sanyo	Daylight	H48	0.2359	2.0246	0.9587	0.8738	0.8636	0.9049
Sanyo	Full	H96	0.1291	2.3001	0.9256	0.8258	0.8753	0.9026
Sanyo	Daylight	H96	0.1929	3.1534	1.3735	1.1337	1.2495	1.2686
Sanyo	Full	H144	0.1181	2.7281	0.8677	0.7753	0.7926	0.8166
Sanyo	Daylight	H144	0.1794	3.6688	1.2940	1.0572	1.1293	1.1427
Hanwha	Full	H12	0.1852	0.4339	0.2136	0.2061	0.2591	0.1756
Hanwha	Daylight	H12	0.2734	0.6286	0.3058	0.2758	0.3032	0.2306
Hanwha	Full	H48	0.1932	1.3501	0.5977	0.5747	0.5864	0.5874
Hanwha	Daylight	H48	0.2856	1.8787	0.8763	0.8169	0.8257	0.8249
Hanwha	Full	H96	0.2035	2.1714	0.8582	0.8626	0.8096	0.8043
Hanwha	Daylight	H96	0.3029	2.8658	1.2691	1.2315	1.1588	1.1364
Hanwha	Full	H144	0.2031	2.5597	0.7860	0.8054	0.7460	0.7431
Hanwha	Daylight	H144	0.3053	3.2725	1.1669	1.1410	1.0609	1.0390
Qcells	Full	H12	0.2805	0.4708	0.9232	0.3274	0.5323	0.4161
Qcells	Daylight	H12	0.4089	0.6817	1.3417	0.4565	0.7484	0.5986
Qcells	Full	H48	0.2806	1.3077	1.1842	0.8076	0.9859	1.0987
Qcells	Daylight	H48	0.4099	1.9054	1.7244	1.1600	1.4200	1.5963
Qcells	Full	H96	0.2553	2.0549	1.2549	1.1368	1.1953	1.3596
Qcells	Daylight	H96	0.3780	3.0413	1.8472	1.6472	1.7385	1.9907
Qcells	Full	H144	0.2436	2.4185	0.9087	0.8436	0.8502	0.7889
Qcells	Daylight	H144	0.3690	3.6631	1.3587	1.2216	1.2407	1.1523

6 Rank sensitivity

7 Independent evidence verification

An independent NumPy/Pandas verifier reads the saved H144 predictions, labels, timestamps, and validity arrays without importing the production metric or mask functions. It re-derives primary and Daily-matched masks, Persistence forecasts, RMSE, MAE, Train-range nRMSE, ranks, and sample counts. All 4,414 comparisons passed the registered numerical tolerances; the maximum absolute discrepancy was 8.51×10^{-12} and the maximum relative discrepancy was 3.23×10^{-11} . The verifier confirmed primary wins of 12, 9, 2, and 1 for the Inverted-variate, Depthwise, Joint-patch, and Discrete implementations, respectively, with zero for

Primary RMSE rank (1 = lowest error)					
	Last-value Persistence	Discrete recurrent decoder	Inverted-variate Transformer	Joint-patch Transformer	Depthwise convolutional TCN
Sanyo H12 Full	5	1	4	3	2
Sanyo H12 Daylight	5	2	3	4	1
Sanyo H48 Full	5	4	2	1	3
Sanyo H48 Daylight	5	4	2	1	3
Sanyo H96 Full	5	4	1	2	3
Sanyo H96 Daylight	5	4	1	2	3
Sanyo H144 Full	5	4	1	2	3
Sanyo H144 Daylight	5	4	1	2	3
Hanwha H12 Full	5	3	2	4	1
Hanwha H12 Daylight	5	4	2	3	1
Hanwha H48 Full	5	4	1	2	3
Hanwha H48 Daylight	5	4	1	3	2
Hanwha H96 Full	5	3	4	2	1
Hanwha H96 Daylight	5	4	3	2	1
Hanwha H144 Full	5	3	4	2	1
Hanwha H144 Daylight	5	4	3	2	1
Qcells H12 Full	3	5	1	4	2
Qcells H12 Daylight	3	5	1	4	2
Qcells H48 Full	5	4	1	2	3
Qcells H48 Daylight	5	4	1	2	3
Qcells H96 Full	5	3	1	2	4
Qcells H96 Daylight	5	3	1	2	4
Qcells H144 Full	5	4	2	3	1
Qcells H144 Daylight	5	4	2	3	1

Figure S1: RMSE ranks for all 24 primary array–horizon–scope combinations. Rank one denotes the smallest error. Short column names correspond to Last-value Persistence, Discrete recurrent decoder, Inverted-variate Transformer, Joint-patch Transformer, and Depthwise convolutional TCN.

Alt text: Heat map of RMSE ranks across 24 array, horizon, and scope combinations. Columns denote Last-value Persistence, Discrete recurrent, Inverted-variate, Joint-patch, and Depthwise TCN; rankings change across conditions.

Last-value Persistence. It also confirmed that Daily Persistence was lower than the post hoc best-of-four neural envelope in 22 comparisons; the two envelope wins were Hanwha H12 full-timeline and daylight.

8 Inference measurement details

Parameter counts include all learned tensors. Batch-one latency and batch-256 throughput were measured in evaluation and inference modes on an NVIDIA GeForce RTX 3060 Laptop GPU with float32 inputs of shape $[B, 72, 17]$. Warm-up preceded synchronized repetitions. Timing excludes model loading, disk input/output, data loading, and host-to-device transfer. Peak allocation is reported in MiB. These hardware-specific measurements support only within-environment comparisons.

9 Unmatched-sample diagnostic

Deleting unavailable lag points from Daily Persistence alone changes its target support and can reverse a headline comparison even when each metric is calculated correctly. The matched-support analysis applies one point mask to Daily Persistence, Last-value Persistence, every neural seed, and the labels. Unmatched-support values are therefore treated only as a methodological diagnostic and never as model-performance evidence.